

Replicant: A Truly Free Version of Android

Replicant is a free and open source mobile operating system based on the Android platform. It aims at replacing proprietary Android apps and components with open source alternatives. It is security focused, as it blocks all known Android backdoors.



Smartphones have evolved from being used just for communicating with others to offering a wide range of functions. The fusion between the Internet and smartphones has made these devices very powerful and useful to us. Android had been a grand success in the smartphone business. It's no exaggeration to say that more than 80 per cent of the smartphone market is now occupied by Android, which has become the preference of most mobile vendors today.

The reason is simple, Android is free and available to public.

But there's a catch. Have you ever wondered how well Android respects 'openness'? And how much Android respects your freedom? If you haven't thought about it, please take a moment to do so. When you're done, you will realise that Android is not completely open to everyone.

That's why we're going to explore Replicant — a truly free version of Android.

Android and openness

Let's talk about openness first. The problem with a closed source program is that you cannot feel safe with it. There have been many incidents, which suggest that people can easily be spied upon through closed source programs.

On the other hand, since open source code is open and available to everyone, one cannot plant a bug in an open source program because the bug can easily be found. Apart from that aspect, open source programs can be continually improved by people contributing to them—enhancing a feature and writing software patches, also there are many user communities that

will help you if you are stuck with a problem.

When Android was first launched in 2007, Google also announced the 'Open Handset Alliance (OHA)' to work with other mobile vendors to create an open source mobile operating system, which would allow anyone to work on it. This seemed to be a good deal for the mobile vendors, because Apple's iPhone practically owned the smartphone market at that time. The mobile vendors needed another player, or 'game changer', in the smartphone market and they got Android.

When Google releases the Android source code to the public for free, it is called 'stock Android'. This comprises only the very basic system. The mobile vendors take this stock Android and tailor it according to their device's specifications—featuring unique visual aspects such as themes, graphics and so on.

OHA has many terms and conditions, so if you want to use Android in your devices, you have to play by Google's rules.

The following aspects are mandatory for each Android phone:

- Google setup-wizard
- Google phone-top search
- Gmail apps
- Google calendar
- Google Talk
- Google Hangouts
- YouTube
- Google maps for mobiles
- Google StreetView
- Google Play store
- Google voice search